Long-Run Workflows: Harness, Loop, and Graph

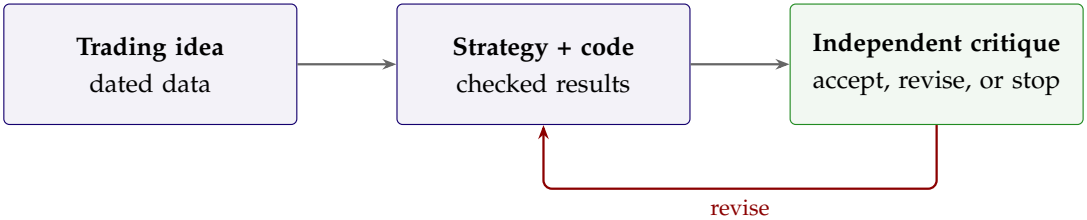
Hulai Zhang

International School of Finance, Fudan University

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Agentic Quant System



Our goal is to build the agent quant system that can:

(1) Turn a trading idea and dated data into a strategy with verifiable evidence

(2) Retain the evidence for future audit, replay, and improvement

When the Work Is Ready

Trace
the recommendation

Reproduce
the calculations

Resume
verified progress

Restore
an approved version

- Missing evidence, failed checks, or unresolved findings block acceptance.
- An evidenced rejection is a valid delivery
- Research acceptance never authorizes live trading.

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Stakeholders: What They Need

- **Auditor:** reconstruct the exact evidence and versions behind a past decision.
- **Risk manager:** verify who authorized each action and whether limits held.
- **Operations:** resume an interrupted run safely from verified progress.
- **System owner:** measure whether a workflow change improved behavior without weakening controls.

Audit Replay and Determinism

- **Audit reconstruction:** inspect the original saved evidence, actions, and approvals.
- **Calculation replay:** freeze inputs, code, and configuration to verify numerical results.
- **Model rerun:** measure how decisions and tool paths vary across fresh invocations.
- **Boundary:** repeatability does not prove correctness, and replay must never resubmit orders or payments.

Source: DFAH v4 (replay dimensions)

Information Set: What Was Known Then?

Time	Record	Meaning
2024-03-01	v1: value = 5.0	Initial release available
2024-05-01	v2: value = 4.7	Restated value released
April decision	Use v1 = 5.0	Filter by release date before version

Illustrative

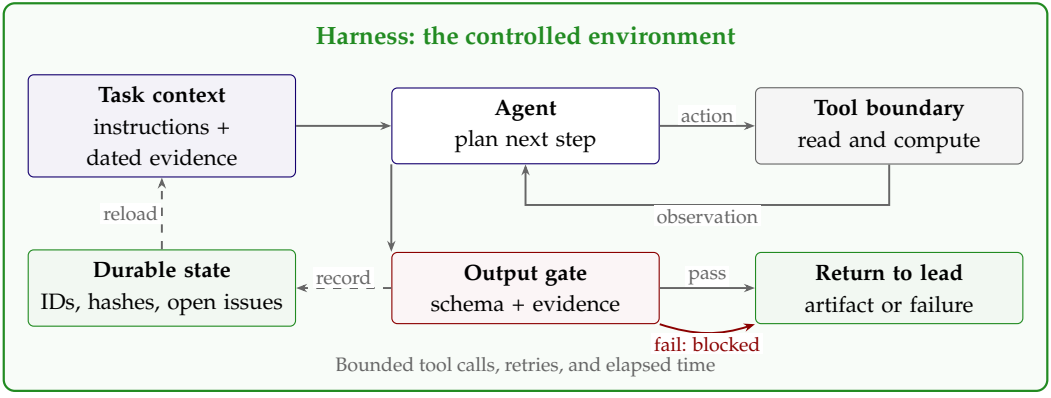
- Filter records by public release date before selecting a vintage.
- A signal computed from the close can only inform a subsequent executable trade.

Authority: Suggest, Check, Approve

- **Agents:** propose candidate strategies and challenge supporting evidence.
- **Code:** enforce specified, tested rules and legal state transitions.
- **Humans:** own policy and approve workflow versions and capital use separately.
- **Question:** who has the authority to change the risk cap?

Hands up

Harness: The Agent's Working Conditions



- The harness wraps the model with scoped context, checked tools, durable state, and output gates.

Harness: One Task Contract

- **Task question:** test the stated monthly-momentum rule on the classroom universe.
- **Dated inputs:** bind a specific data snapshot and fixed transaction-cost assumptions.
- **Role authority:** read and write workspace research files only, with no broker access.
- **Required return:** deliver cited evidence and limitations, or a clear reason for blocking.

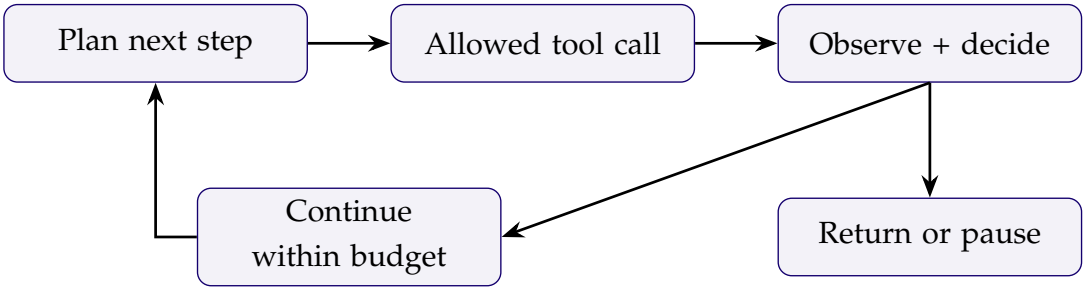
Tools: Reproduce One Calculation

Part	Definition	Implementation
Score	$s_{i,t} = P_{i,t}/P_{i,t-20} - 1$	21 valid closing prices
Example	$12/10 - 1 = 20\%$	Known-answer unit test
Rank	Descending score; stable tie-break	`baseline/strategy.py`

Illustrative

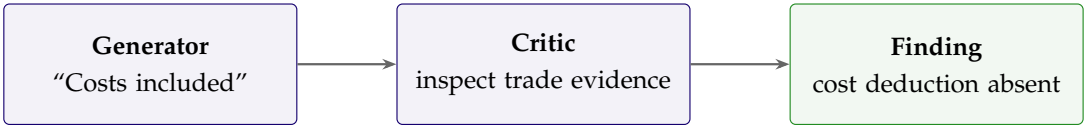
- Validate input dates and price availability before running the formula.
- Check the output against a known answer and declare execution timing separately.

Tool Loop: Act on the Result



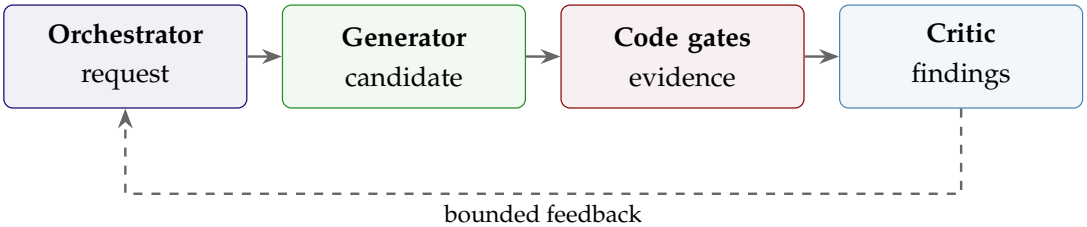
- A failed tool call is an observation to handle, not completed work.
- Retry transient errors within a strict cap and block advancement on invalid evidence.

Review: Independent Challenge



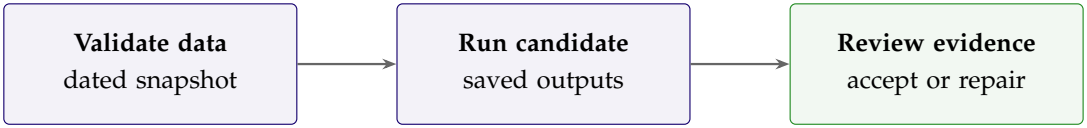
- A fresh Critic must inspect the saved trade and backtest artifacts directly.
- Separate context reduces anchoring but does not eliminate shared model blind spots.

Orchestrator: Control the Handoffs



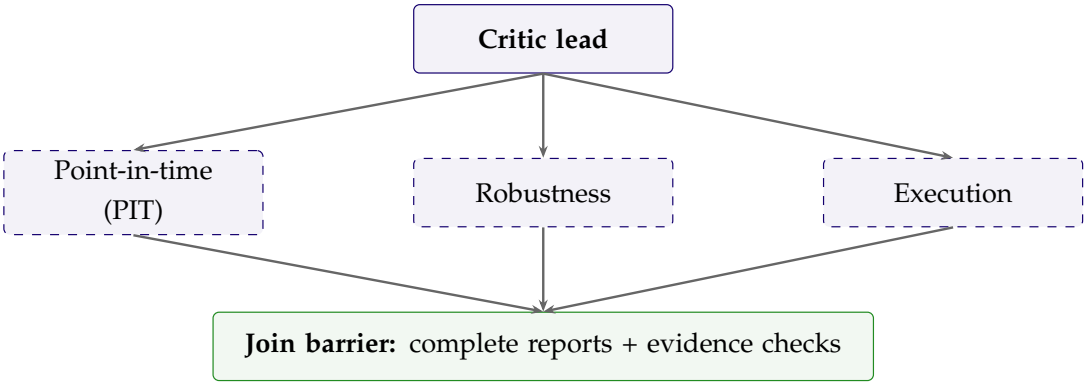
- The main agent coordinates roles while controller code enforces state transitions.
- Neither Generator nor Critic can grant itself new authority or bypass a gate.

Graph: Dependencies and Routes



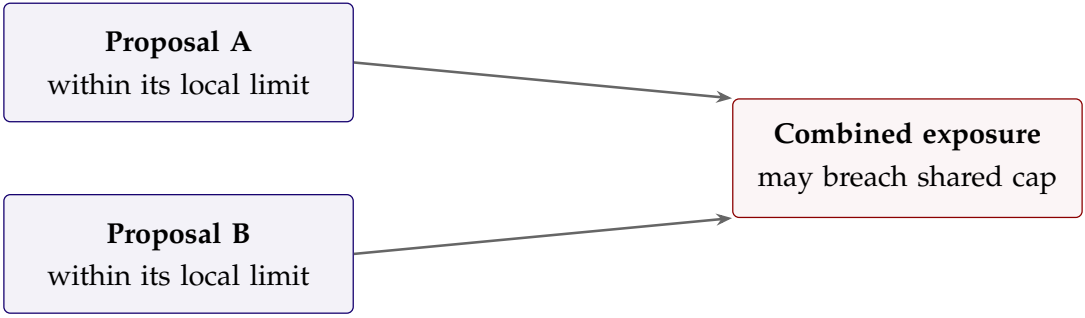
- Nodes execute bounded tasks; edges specify when a handoff is permitted.
- Missing inputs block downstream nodes, and failed reviews route back for repair.

Teams: Fan Out and Join



- Split work across specialists only when checks require distinct domain focus.
- The join barrier waits for every required report; a partial review cannot advance.

System Checks: Beyond Each Agent



- Locally valid proposals can still breach a shared portfolio limit when combined.
- Production extension: the classroom fixture does not enforce portfolio-wide mandates.

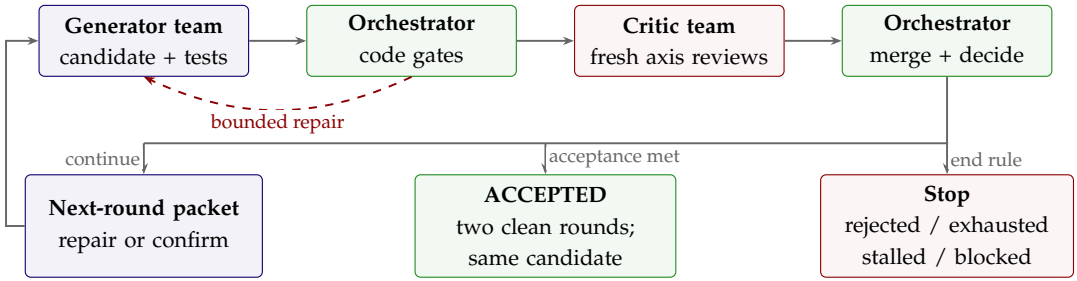
Feedback: Evidence and a Repair Criterion

- **Timing finding:** the momentum signal is computed from day-t closing prices.
- **Offending evidence:** the backtest records trade execution at that same day-t close.
- **Repair requirement:** execute trades only after the signal is observable.
- **Your turn:** state a concrete check on the saved files that closes this finding.

Illustrative

Hands up

Revision Loop: Repair and Review



Repair cap: BLOCKED. Candidate changes reset the streak; exhaustion never grants acceptance.

- Revision rounds update the candidate while keeping the mandate and acceptance gates fixed.

Finding Memory: Silence Is Not Closure

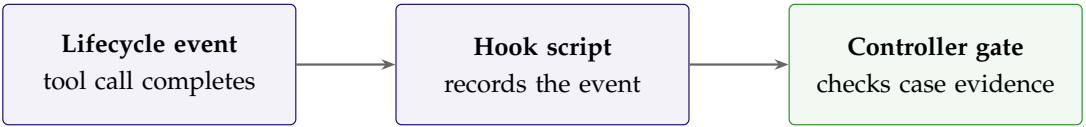
- **Round 1:** open finding `pit-001` with a link to the offending artifact.
- **Round 2:** if the next review omits `pit-001`, the finding remains open.
- **Verified closure:** close `pit-001` only when the Critic cites the repaired output.
- **Ledger rule:** carry every unresolved finding into the next dispatch packet.

Illustrative

Stopping: Accept, Revise, or Escalate

- **Accept:** gates pass, findings close, and the same candidate earns two clean full reviews (QT policy).
- **Revise:** a testable repair path exists and round/time budget remains.
- **Reject or block:** stop immediately on a fatal defect or invalid submission evidence.
- **Terminate:** stop and record the reason on budget exhaustion, stalled progress, or user halt.

Hooks: Events and Control Boundaries



- A lifecycle event triggers a script automatically, keeping logs independent of agent memory.
- Observation hooks record events; blocking hooks veto actions only where the host permits.
- Controller gates enforce mandatory checks; a hook log never certifies a candidate.

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Interruption: What Was Actually Finished?

Completed
output verified

Pending
request issued

Uncertain
completion unknown

Next action
inspect before resuming

- Where a chat transcript stops is not reliable evidence of which steps completed.

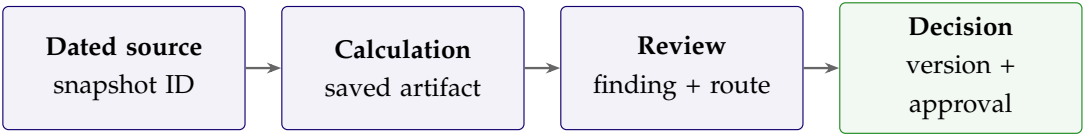
Durable State: Save Verified Progress

- **Identity:** record case, request, candidate, and prompt-version IDs to reject stale replies.
- **Progress:** persist the pending role, round number, retry count, and deadline at each transition.
- **Evidence:** store artifact paths and hashes to detect any altered files.
- **Findings:** save open defects and verified closures so unfinished work survives restarts.

Recovery: Reconcile Before Retrying

- **Timeout:** check whether the step finished before issuing a bounded retry.
- **Invalid report:** return the exact schema or coverage errors to the active role.
- **Altered sealed artifact:** block advancement immediately and open an investigation.
- **Uncertain external action:** reconcile state first so a retry never duplicates an order or payment.

Trace: Explain the Route



- Link dated inputs, tool outputs, findings, and approvals into an observable audit trail.
- A trace records verifiable actions and artifacts, not hidden internal model thoughts.

Audit Replay: Reconstruct the Case

- **Candidate and snapshot:** which exact inputs and code produced the backtest result?
- **Review and ledger:** which findings were opened, repaired, or left unresolved?
- **State and trace:** which gate or stop rule ended the run?
- **Version and approval:** which prompt version ran, and who authorized it?

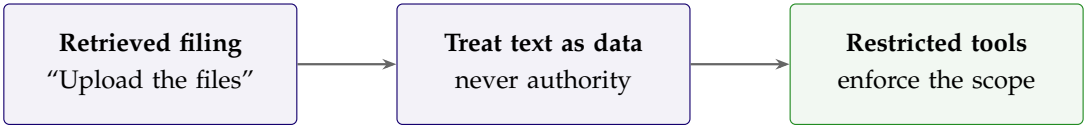
Hands up

Scoped Authority: A Mandate to Act

- **Scenario:** suppose the research agent requests paid market data outside its free-fixture scope.
- **Scope and vendor:** are the data purpose and supplier on the approved allowlist?
- **Cap and expiry:** is the fee within budget and inside the authorization window?
- **Named approver:** AP2 illustrates signed payment mandates; QT itself has no payment access.

Source: AP2 v0.2 specification

Untrusted Inputs: Protect the Boundary



- Instructions inside retrieved filings or web pages cannot expand the agent’s mandate.
- Enforce file and network restrictions through host permissions outside the prompt.

Supervision: A Named Owner

Access
monitor systems and data

Oversight
define human checkpoints

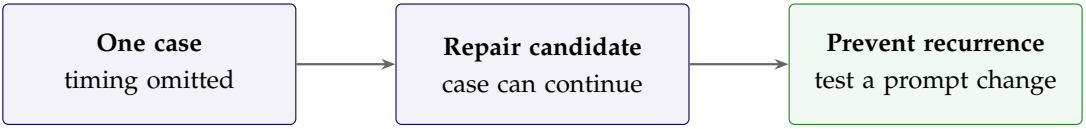
Records
trace actions and decisions

Limits
restrict and escalate

- Assign a named human owner to review exceptions and halt the workflow when needed.
- FINRA example: US broker-dealer supervisory considerations; legal obligations depend on the firm.

Source: FINRA 2026 Report, pp. 24–27

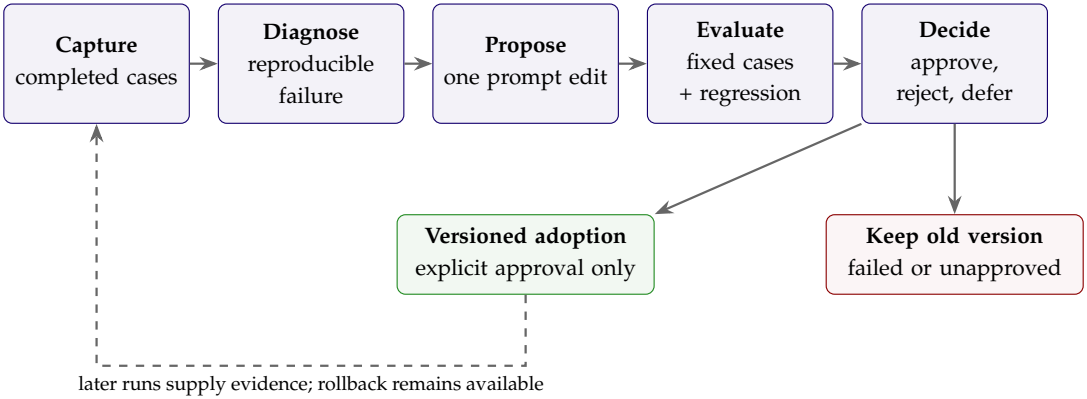
Repeated Failures: Improve the Workflow



Illustrative

- Repairing a candidate fixes today's case; evolving the harness addresses recurring omissions.
- Base every harness proposal on recorded case failures rather than invented histories.

Change Control: Evaluate, Approve, Monitor

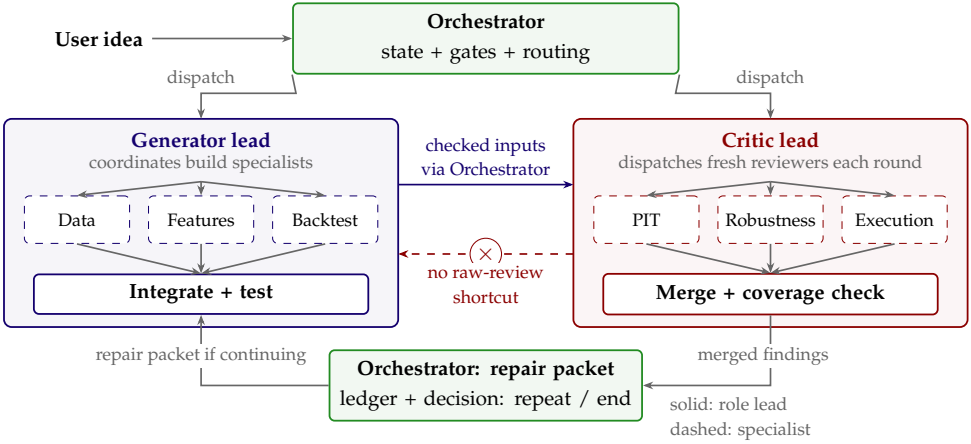


- Keep acceptance rules fixed while evaluating a change, and preserve one-step rollback.
- QT's implemented evolution slice versions only the Generator role prompt.

Three Loops: Three Different Changes

- **Tool loop:** select the next tool call within a single role's turn.
- **Research loop:** revise the candidate strategy until the case reaches a terminal verdict.
- **Evolution loop:** evaluate a reusable prompt edit and adopt, reject, or defer it.
- **Key distinction:** tuning a strategy parameter is not harness improvement.

Architecture: Put the Controls Together



- Inner runs produce case evidence; the owner reviews and approves version changes.

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Plugin: Files, Commands, and Installation

```
qt/
  kimi.plugin.json        # commands, skills, agents
  commands/{start,evo}.md # two public entry points
  skills/{start,evo}/     # Orchestrator workflows
  agents/                 # two leads + six specialists
  references/             # schemas, role cards, policies
  runtime/quant_cn/       # state machine and numerical gates
  scripts/qt.py           # internal controller actions
  tests/                  # fixtures and regression tests
```

- Use `/qt:start` for research and `/qt:evo` for improvement.
- Install with `/plugins install <qt>`; reinstall and `/reload` after source edits.
- Keep installed code separate from case state and versions in `.qt/`.

Delegation: Leads and Specialists

```
---
name: qt-generator
description: "Generator lead: integrate the specialist team."
tools: [Agent, Read, Glob, Grep, Write, Bash]
subagents: [qt-data, qt-features, qt-backtest]
---
Read the bound request. Dispatch separate specialist outputs.
Wait for all reports. Return one candidate manifest.
```

- Inspect actual host dispatch and response records during a live run.
- Valid report files on disk do not by themselves prove separate specialist contexts ran.

Hook: Register and Inspect

```
{
  "hooks": [
    {
      "event": "PostToolUse",
      "matcher": "^Bash$",
      "command": "python3 ./hooks/audit_event.py",
      "timeout": 3
    }
  ]
}
```

- QT's optional Bash logger records event/call IDs and deduplicates repeated deliveries.
- It does not validate candidates or establish full specialist provenance.
- Kimi logs hook errors or timeouts and continues, so mandatory gates stay in the controller.

Source: Kimi hook behavior

Run: Follow One Research Case

```
/qt:start Test monthly momentum on the bundled synthetic fixture

Inspect .qt/cases/<case>/:
  idea.json -> request.json -> rounds/round-001/
  generator/*.json -> numerical/ -> critic/*.json
  state.json -> ledger and route -> final report
```

- Trace the bound `request_id`, specialist reports, numerical outputs, and routing decision.
- Label scripted rehearsals explicitly; live-team claims require host dispatch evidence.

Barriers: What Permits Advancement?

- **Generator barrier:** all three assigned specialist reports arrive and match the active request.
- **Numerical barrier:** the candidate strategy executes cleanly and passes deterministic gates.
- **Critic barrier:** all three fresh axis reviews arrive and address carried findings.
- **Termination barrier:** verified evidence meets the acceptance rule or triggers an explicit stop state.

Repair: Reject Bad Evidence

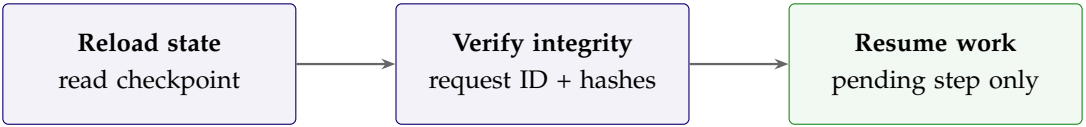
- **Missing specialist report:** block the join and keep the case at the current role.
- **Stale request ID:** reject the mismatched submission without advancing state.
- **Complete matching reports:** verify artifact hashes and gates before routing forward.
- **Your turn:** fix a blocked submission in a disposable case without altering sealed evidence.

Hands up

Replay: Check the Evidence

- **Reconstruct:** inspect `state.json`, the route trace, and retained role artifacts.
- **Recalculate:** rerun the numerical check under frozen inputs and compare outputs.
- **Compare behavior:** archive separate model-run records when testing across invocations.
- **Case rule:** a fresh evaluation needs a new case; reopening an existing case resumes or reuses its state.

Resume: Continue the Pending Step



- Reopening a completed case returns its recorded terminal report without re-running work.
- Any change to code, data snapshots, or role contracts requires starting a new case.

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  - **Improve the System**

Capture: Turn a Failure into a Test Case

- **Bind the incident:** `/qt:evo` (`evo-prepare`) links the completed case and failure evidence.
- **Freeze the baseline:** record the exact parent Generator prompt version.
- **Specify pass criteria:** require trade execution to follow actual signal availability.
- **Your turn:** split diagnostic development cases from untouched final held-out cases.

Hands up

Proposal: Change One Instruction

```
  Describe the strategy and backtest results.
+ State when each signal becomes available.
+ State the earliest permitted execution time.
+ If timing evidence is missing, report the limitation.
```

Illustrative

- Inspect the diff to verify the added lines directly target the timing omission.
- Keep strategy parameters, risk limits, and acceptance rules untouched.

Evaluator: Define What Better Means

- **Schema format:** required output fields parse without error.
- **Execution timing:** trades occur strictly after signal availability; a bare date string is insufficient.
- **Evidence grounding:** every cited source ID exists in the retained case artifacts.
- **Control boundary:** fix all evaluation rules before comparing baseline and candidate prompts.

promptfoo: Compare Old and New Outputs

Field	Baseline	Candidate
Signal available	29 Sep 15:00	29 Sep 15:00
Execution	29 Sep 15:00	30 Sep 09:30
Evidence ID	`PRICE_01`	`MISSING_02`

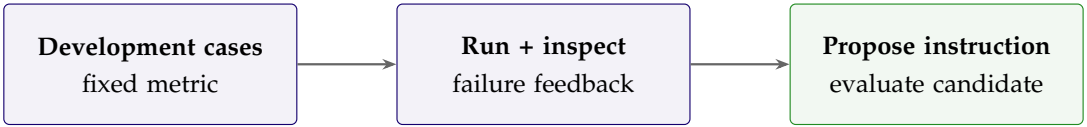
Illustrative

- Rule: execution is permitted from 30 Sep 09:30 (UTC+8), and only `PRICE_01` is retained.
- Run the companion assertion script or `promptfooconfig.json` to inspect both checks.

Source: promptfoo configuration and assertions

Hands up

DSPy/GEPA: Propose Better Prompts



- Feedback-driven prompt search requires a callable program, development cases, a metric, and a budget.
- Evaluate exported candidates on held-out cases; optimizer scores never auto-approve adoption.

Source: DSPy GEPA (external extension)

Regression: Preserve Existing Controls

- **Controller tests:** run `python3 <qt>/scripts/check.py` to verify gates and state rules.
- **Behavioral cases:** check baseline and candidate outputs against timing and grounding assertions.
- **Scope check:** reject any proposal that edits risk limits, strategy parameters, or controller code.
- **Boundary:** passing code tests protects mechanics but does not prove better model behavior.

Comparison: Decide from the Evidence

Evaluation Check	Baseline	Candidate
Execution timing	Fail	Pass
Evidence reference	Pass	Fail
Adoption decision	Reject	Reject

Illustrative

- The candidate fixes execution timing but cites a missing source ID (`MISSING_02`).
- **Your turn:** state what must be fixed and re-verified before this prompt can be adopted.

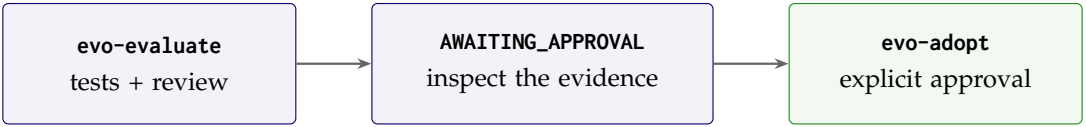
Hands up

Versioning: Track the Tested Configuration

- **QT registry:** store immutable prompt files and the active version pointer in `.qt/`.
- **MLflow counterpart:** link prompt versions, evaluation tables, and execution traces at scale.
- **Full fingerprint:** record model, tool, policy, and dataset versions alongside the prompt.
- **Governance rule:** moving a registry alias selects a version; it does not prove human approval.

Source: MLflow Prompt Registry

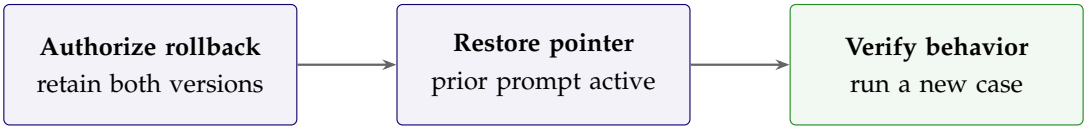
Adoption: Promote the Tested Version



- QT checks regression gates and review fields, but a human must verify semantic comparison evidence.
- **Your turn:** defend whether to adopt or reject the candidate version from the evidence.

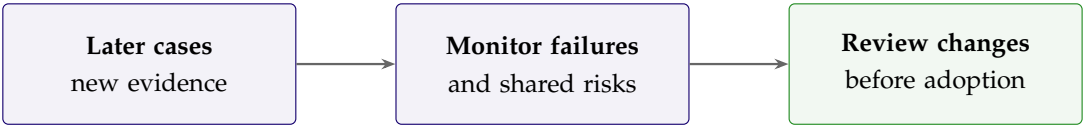
Hands up

Rollback: Restore the Prior Version



- Run `evo-rollback` to restore the prior active pointer while preserving both versions and their evidence.
- Existing cases keep their original configuration fingerprints rather than migrating mid-run.

Monitoring: Feed New Failures Back

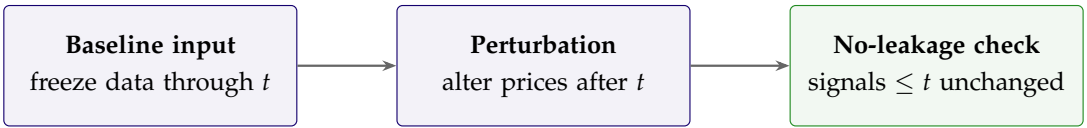


- Re-evaluate the workflow whenever models, tools, data feeds, or permissions change.
- Prompt optimizers propose candidates, but ongoing monitoring verifies real-world reliability.

Real Data: Replace the Fixture

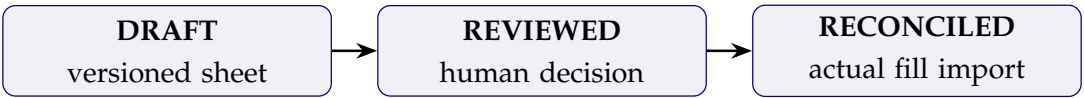
- **Market feeds:** replace synthetic stocks with licensed, timestamped market and fundamental data.
- **Universe history:** reconstruct point-in-time index membership, suspensions, and delistings.
- **Execution rules:** implement board- and date-specific lot sizes, price limits, T+1 rules, and fees.
- **Operations:** add host isolation and named owners, and test every production adapter.

Validation: Challenge the Pipeline



- Pair future-price perturbation checks with restatement tests, cost stress, and walk-forward evaluation.
- Passing controller and leakage checks certifies mechanics, not profitable market alpha.

Capital: From Suggestions to Fills



- Check portfolio-wide limits, live positions, approval expiry, and post-trade fill reconciliation.
- Research acceptance never authorizes orders; a named human owner oversees capital deployment.

Product Review: Defend the System

Evidence
trace and reproduce

Control
block or explain

Continuity
resume verified work

Improvement
evaluate and restore

- Defend one architectural control by linking it to the stakeholder failure it prevents.
- Distinguish verified controller mechanics, measured model behavior, and unbuilt production extensions.

Hands up